

46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: JEE APEX & SR IIT MODEL-A **DT: 28 / 12 /2025**
Time: 3 Hrs **GTM - 05** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. An electron enters symmetrically between two oppositely charged parallel plates, each of length $15\,cm$, with a horizontal velocity of $v_x = 2.0\times10^6\ m/s$. If the electric field between the plates is $7.5\,V/cm$, then the vertical component of velocity of the electron, while leaving the field region is__ $\left(m_e = 9.1\times10^{-31}kg,\ e = 1.6\times10^{-19}C\right)$ (nearly)

A) $3.0\times10^7\ m/s$ B) $2.0\times10^7\ m/s$

C) $1.0\times10^7\ m/s$ D) $4.0\times10^5\ m/s$

For More Join: @IITJEE_Advanc

2. Given the two statements:

Statement – I : The equivalent emf of two non-identical batteries connected in series is always greater than the emf of either battery.

Statement – II : The equivalent internal resistance of two non-ideal batteries connected in series is equal to the sum of their internal resistances.

Choose the correct answer:

A) Both Statement – I and II are false

B) Statement – I is false, Statement – II is true

C) Both Statement – I and II are true

D) Statement – I is true, Statement – II false

3. A uniform circular disc of radius R and mass M is rotating about an axis perpendicular to its plane and passing through its centre. A smaller disc of radius $\frac{R}{3}$ is cut out such that its centre lies at a distance $\frac{2R}{3}$ from the main disc's centre along a diameter. Find the moment of inertia of the remaining part of original disc about the axis as given above.

A) $\frac{35}{54}MR^2$ B) $\frac{24}{54}MR^2$ C) $\frac{31}{54}MR^2$ D) $\frac{29}{54}MR^2$

4. A $2g$ ice cube at $-5°C$ is heated until it becomes steam at $120°C$.

Find the heat required for this process.

(Given: $s_i = 2100$; $s_w = 4200$; $s_s = 2000$; $L_f = 3.3\times10^5$; $L_v = 2.25\times10^6$. every value is in SI unit)

A) $6000\ J$ B) $6600\ J$ C) $6400\ J$ D) $6100\ J$

5. In a hydrogen – like atom, the orbital radius in the ground state is $5.2\times10^{-11}m$. In some excited orbit, the radius is $4.68\times10^{-10}m$. Find the ratio of de-Broglie wavelengths of excited state to ground state $\left(\frac{\lambda_{excited}}{\lambda_{ground}}\right)$

A) 6 B) 9 C) 4 D) 3

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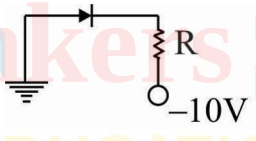
6. A bob of mass m tied with a string of length l is given a horizontal velocity at the lowest point such that $v_0 = \sqrt{6gl}$. It now moves in a vertical circle. Let B be the point where the string makes an angle $45°$ with downward vertical, and C be the point where it makes $135°$ with downward vertical. Find the ratio of kinetic energies at points B and C $(K_B : K_C)$ (nearly)

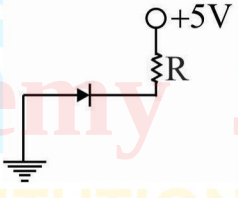
A) $1:2$ B) $2:1$ C) $3:1$ D) $4:1$

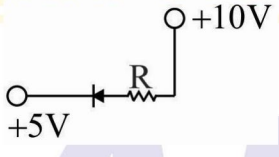
7. Given is a thin plano convex lens of glass (refractive index μ) having radius of curvature R. Curved side is polished for complete reflection. At what distance from the lens, an object be placed on the optic axis so that the image gets formed on the object itself?

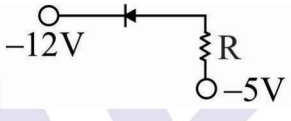
A) $\frac{R}{\mu}$ B) $\frac{R}{(2\mu-3)}$ C) $\frac{R}{2\mu}$ D) $\frac{R}{(2\mu-1)}$

8. In the following, which one of the diodes is reverse biased?

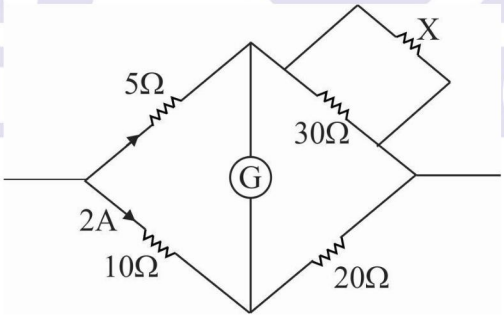
A) 

B) 

C) 

D) 

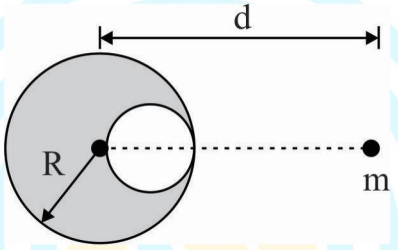
9. In the given circuit if current through the galvanometer is zero, then current through $'X'$ is



A) $4A$ B) $\frac{2}{3}A$ C) $\frac{4}{3}A$ D) $\frac{8}{3}A$

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10. A spherical hollow cavity is made in a lead sphere of radius R, such that its surface touches the outside surface of the lead sphere and passes through its centre. The mass of the sphere before hollowing was M. With what gravitational force will the hollowed-out lead sphere attract a small sphere of mass 'm', which lies at a distance d from the centre of the lead sphere on the straight line connecting the centres of the spheres and that of the hollow, if $d = 2R$:



A) $\frac{7GMm}{18R^2}$ B) $\frac{7GMm}{36R^2}$ C) $\frac{7GMm}{9R^2}$ D) $\frac{7GMm}{72R^2}$

11. A closed organ pipe of length L and an open organ pipe contain gases of densities ρ_1 and ρ_2 respectively. The compressibility of gases are equal in both the pipes. Both the pipes are vibrating in their first overtone with the same frequency. The length of the open pipe is _____

A) $\frac{4}{3}L\sqrt{\frac{\rho_1}{\rho_2}}$ B) $\frac{2}{3}L\sqrt{\frac{\rho_1}{\rho_2}}$ C) $\frac{4}{3}L\sqrt{\frac{\rho_2}{\rho_1}}$ D) $\frac{2}{3}L\sqrt{\frac{\rho_2}{\rho_1}}$

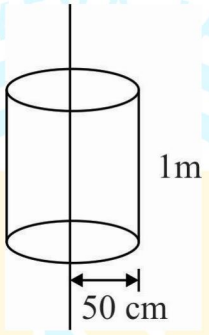
12. If ε_0 is the permittivity of free space and E is the electric field, then $\varepsilon_0 E^2$ has the dimensions:

A) $\left[M^0L^{-2}TA\right]$ B) $\left[ML^{-1}T^{-2}\right]$

C) $\left[M^{-1}L^{-3}T^4A^2\right]$ D) $\left[ML^2T^{-2}\right]$

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13. Electric charge is uniformly distributed along a straight wire of radius $1\ mm$. The charge per centimeter length of the wire is Q coulomb. Another cylindrical surface of radius $50\ cm$ and length $1\ m$ symmetrically encloses the wire as shown in figure. The total electric flux passing through the cylindrical surface is



A) $\frac{100Q}{\varepsilon_0}$ B) $\frac{10Q}{\varepsilon_0}$ C) $\frac{100Q}{\pi\varepsilon_0}$ D) $\frac{Q}{\varepsilon_0}$

14. Given below are two statements:

Statement I : In a vernier calipers, one Vernier scale division is always smaller than one main scale division.

Statement II: The vernier constant is given by one main scale division multiplied by the number of Vernier scale divisions

A) Statement I is true but Statement II is false

B) Statement I is false but Statement II is true

C) Both Statement I and Statement II are false

D) Both Statement I and Statement II are true

15. The work functions of sodium (Na) and potassium (K) metals are $2.3\ eV$ and $2.0\ eV$, respectively. If monochromatic light of wavelength $480\ nm$ is incident on each metal surface, photoelectric emission will occur from: (Take $hc = 1240\ eVnm$).

A) Both Na and K B) Neither Na nor K

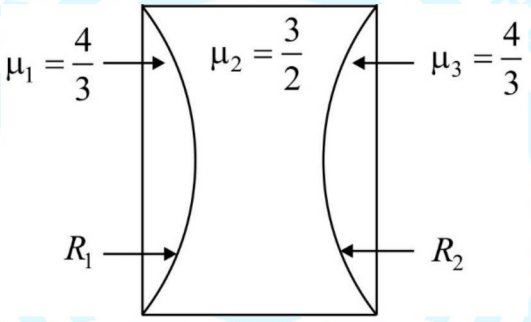
C) Na only D) K only

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16. Two spherical bodies made of identical material have radii $0.3\ m$ and $0.9\ m$, placed in the same environment. The smaller sphere is at $700\ K$, and the larger sphere is at $350\ K$. If the power radiated by the smaller sphere is P, then the power radiated by the larger sphere is:

A) $4P$ B) $\frac{9P}{16}$ C) $64P$ D) $16P$

17. Find the equivalent power of the thin lens combination shown in the figure.



A) $+\left(\frac{R_1 + R_2}{3R_1R_2}\right)$ B) $-\left(\frac{R_1 + R_2}{3R_1R_2}\right)$

C) $-\left(\frac{R_1 + R_2}{6R_1R_2}\right)$ D) $+\left(\frac{R_1 + R_2}{6R_1R_2}\right)$

18. **Assertion (A):** In Young's double-slit experiment, if the entire apparatus is immersed in water, the fringe width decreases.

Reason (R) : The wavelength of light reduces inside a medium of refractive index greater than 1.

Choose the correct option:

A) Both (A) and (R) are true but (R) is not the correct explanation of (A)

B) Both (A) and (R) are true and (R) is the correct explanation of (A)

C) (A) is true but (R) is false

D) (A) is false but (R) is true

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19. A parallel-plate capacitor of capacitance $30\mu F$ is connected to a $200\,V$ battery. A dielectric slab of dielectric constant $K=3$ completely fills the space between the plates. After inserting the slab, the extra charge stored and the change in energy of the capacitor are:

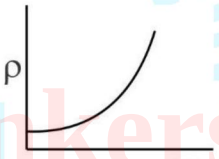
A) $12\,mC$ and $1.2\,J$ B) $18\,mC$ and $6\,J$

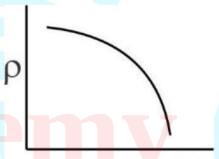
C) $6\ mC$ and $2\ J$ D) $24\ mC$ and $8\ J$

20. Which of the following graph represents the variation of resistivity (ρ) with temperature (T) for copper?

A) 

B) 

C) 

D) 

SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

21. A driver sitting inside the parked car looks in a convex side-mirror of radius of curvature $R=2m$. A car behind approaches with speed $72\,kmph$. When the car is at a distance $4\,m$ from the mirror, the magnitude of the acceleration (in $\mathrm{m/s^2}$) of its image in the mirror is a. Find the integer value of $10\,a$.

22. Two soap bubbles of radii $r_1=3\,cm$ and $r_2=6\ cm$ are in contact. What is the radius of curvature R (in cm) of the common surface?

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23. Two particles A and B each of mass $1\,kg$ have position vectors $\vec{r}_A = (2t)\hat{i} + (-t)\hat{j}$ and $\vec{r}_B = (t)\hat{i} + (2t)\hat{j}$ (in meters). At $t = 1s$ find the magnitude of angular momentum (in $\text{kgm}^2\text{s}^{-1}$) of particle A about the position of particle B.

24. Three thermal conductors of same length and area of cross section and having thermal conductivities $k, 2k, 3k$ respectively are joined in series. If the temperature difference across the combination of three conductors is $110^0 C$, then the temperature difference across the middle conductor in $\left({}^0C\right)$ is ____

25. A particle is projected at angle $30°$ to the horizontal with a speed $30\ m/s$. Let h_0 be the vertical height gained in the first second, and h_1 be the height gained in the last second before reaching the maximum height. The value of $\frac{h_0}{h_1}$ is___. (Take $g = 10\ m/s^2$)

CHEMISTRY MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. The radius of the 2nd orbit of Li^{2+} is x. The expected radius of the 3rd orbit of Be^{3+} is

A) $\frac{16}{27}x$ B) $\frac{4}{9}x$ C) $\frac{9}{4}x$ D) $\frac{27}{16}x$

27. Select the correct statements from the following

i) Cyclooctene can show geometrical isomerism

ii) Trans-2-butene has higher melting point than cis-2-butene

iii) Maleic acid and fumaric acids are geometrical isomers

iv)Cis-1,2-dichloroethene has lower dipole moment than trans-1,2-dichloroethene

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A) i), ii), iii) only
B) i), ii) only
C) ii), iii), iv) only
D) i), ii), iii), iv)

28. Choose the correct option for free expansion of an ideal gas under adiabatic condition is

A) $q = 0,\ W = 0,\ \Delta T = 0$
B) $q = 0,\ W = 0,\ \Delta T \neq 0$
C) $q = 0,\ W = 0,\ \Delta T < 0$
D) $q = 0,\ W = 0,\ \Delta T > 0$

29. Which of the following is correct increasing order of electronegativity of the elements

A) C, N, Si, P
B) N, Si, C, P
C) Si, P, C, N
D) P, Si, N, C

30. Which of the following Lanthanide ions are Diamagnetic

A) Nd^{3+} & Eu^{3+}
B) Lu^{3+} & Ce^{4+}
C) Nd^{3+} & Ce^{4+}
D) Lu^{3+} & Eu^{3+}

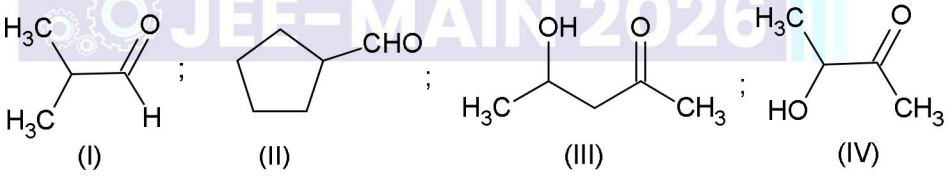
31. **Statement (I):** One mole of 1-butyne reacts with excess of sodium to liberate one mole of H_2 gas.

Statement (II): 5.4g of 1-butyne reacts with excess CH_3MgBr to liberate 2.24L of CH_4 at STP.

In the light of the above statements, choose the most appropriate answer from the options given below:

A) Statement I is false but Statement II is true
B) Both Statement I and Statement II are true
C) Statement I is true but Statement II is false
D) Both Statement I and Statement II are false

32. Which of the following compounds can give positive Fehling's test



(I) ; (II) ; (III) ; (IV)

A) (I), (II) and (III) only
B) (II), (III) and (IV) only
C) (I), (II) and (IV) only
D) (I), (III) and (IV) only

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33. Which of the following concentrated aqueous solutions on electrolysis using inert electrodes liberate oxygen at anode and hydrogen at cathode.

A) $MgSO_4$ B) KBr C) CH_3COONa D) $AgNO_3$

34. Statement – I: Benzyl chloride shows high reactivity towards the SN^1 reaction

Statement – II: chloromethyl cyclohexane shows high reactivity towards SN^2 reaction

A) Statement I is false but Statement II is true

B) Both Statement I and Statement II are true

C) Statement I is true but Statement II is false

D) Both Statement I and Statement II are false

35. Which of the following compounds is not a vitamin?

A) Ascorbic acid B) Pyridoxine C) Tryptophan D) Thiamine

36. Match list-I with list-II

List-I		**List-II**	
1)	Tl > In > Al > Ga > B	p)	Density
2)	B < Al < Ga < In < Tl	q)	Atomic radius
3)	B > Al > Ga > In > Tl	r)	Melting point
4)	B > Al > Tl > In > Ga	s)	Boiling point

A) 1)-q : 2)-r : 3)-s : 4)-p B) 1)-r : 2)-p : 3)-s : 4)-q

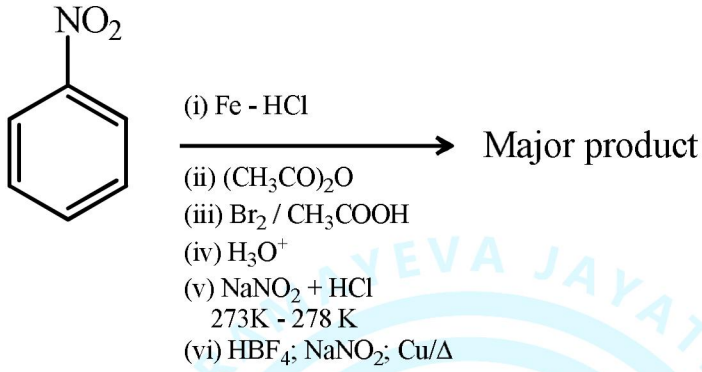
C) 1)-p : 2)-q : 3)-s : 4)-r D) 1)-q : 2)-p : 3)-s : 4)-r

37. Sucrose decomposes in acid solution into glucose and fructose according to 1st order rate law, with half life $(t_{1/2} = 3\,\text{hours})$. The fraction of sample of sucrose remains after 8 hours is (given antilog of 0.8024=6.345)

A) 0.158 B) 6.345 C) 0.8024 D) 0.231

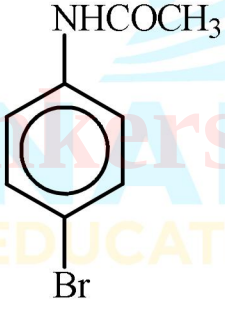
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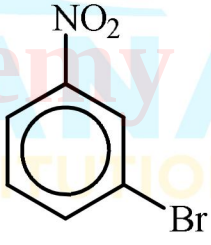
38.



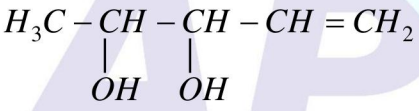
A) 

B) 

C) 

D) 

39. How many stereoisomers are possible for the given molecule?



A) 4 B) 8 C) 6 D) 10

40. Consider the following equation:

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g), \Delta H = -190kJ$, which will increase the yield of SO_3 at equilibrium from the following:

A) Increasing temperature B) Decreasing pressure

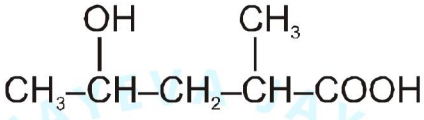
C) Adding more SO_2

D) Addition of inert gas at constant volume

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41. During electrolysis of acidified water, O_2 gas is formed at the anode. To produce O_2 gas at the anode with 100% efficiency at the rate of 0.224 cc per second at STP, current passed is

A) 0.224 amp B) 2.24 amp C) 9.65 amp D) 3.86 amp

42. The IUPAC name of  is:

A) 1-Hydroxy-4-methylpentanoic acid

B) 1-Methyl-3-hydroxypentanoic acid

C) 4-Hydroxy-2-methylpentanoic acid D) 4-Carboxypentanol-2.

43. Which of the following orders is correct about the Δ_0 values?

A) $[Co(CN)_6]^{3-} > [Co(H_2O)_6]^{2+}$ B) $[Co(H_2O)_6]^{2+} > [Co(H_2O)_6]^{3+}$

C) $[Co(H_2O)_6]^{3+} > [Rh(H_2O)_6]^{3+}$ D) $[Co(NH_3)_6]^{3+} < [CoF_6]^{3-}$

44. The correct order of boiling points of the following aqueous solutions is? Assume 100% ionisation.

(1) 0.01 M Na_2SO_4 (2) 0.01 M $Al_2(SO_4)_3$

(3) 0.015 M urea (4) 0.02 M glucose

A) 3 < 1 < 2 < 4 B) 3 < 4 < 1 < 2 C) 2 < 1 < 4 < 3 D) 1 < 3 < 2 < 4

45. Choose the correct set of hybridisation and magnetic behaviour of following complexes?

A) $\left[Ni\left(H_2O\right)_6\right]^{2+}$ d^2sp^3, diamagnetic B) $\left[PtC\ell_6\right]^{2-}$ sp^3d^2, paramagnetic

C) $\left[Ni\left(CO\right)_4\right]$; dsp^2, diamagnetic D) $\left[FeF_6\right]^{4-}$, sp^3d^2, paramagnetic

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SECTION – II
(Numerical Value Answer Type)

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46. Among the following species, the number of species having dsp^2 hybridisation is $'x'$ and the number of species having sp^3d^2 hybridisation is $'y'$ so the value of $(x+y)$ is_____

1) $\left[Ni(CN)_4\right]^{-2}$ 2) SF_4

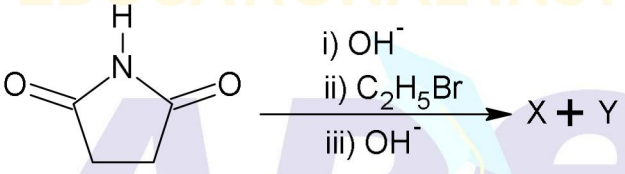
3) $\left[Ni(Cl)_4\right]^{-2}$ 4) $\left[Pt(Cl)_4\right]^{-2}$

5) PF_5 6) PCl_5 7) BrF_5

8) $\left[Co(NH_3)_6\right]^{+3}$ 9) SF_6 10) ClF_3

47. The rate of reaction gets doubled when the temperature changes from 7°C to 17°C. Assuming E_a to remain constant, by what factor will the rate increase when temperature increases from 17°C to 27°C. (Nearest integer)

48. Consider the following sequence of reactions;



If 'X' is succinate ion, the molecular weight of 'Y' is?

49. 4.3g of saturated hydrocarbon is ($C_n H_{2n+2}$) burnt completely and all CO_2 were absorbed in aq KOH solution. A 300 ml 2N KOH solution was required. What is value of 'n'

50. When 0.15 g of an organic compound was analysed using Carius method for estimation of bromine, 0.2397 g of AgBr was obtained. The percentage of bromine in the organic compound is _(Round off nearest integer)_____. (Atomic mass: Silver = 108, Bromine = 80)

For More Join: @IITJEE_Advanc

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

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51. Let $a_1, a_2, a_3 \ldots$ be in G.P having all of its terms positive. If $\sum_{n=1}^{100} a_{2n+1} = 200$, $\sum_{n=1}^{100} a_{2n} = 100$ then the value of $\sum_{n=1}^{100} a_n$ is equal to___

A) 300 B) 150 C) 175 D) 225

52. If solution of the differential equation $3\frac{dy}{dx} + \frac{2y}{x+1} = \frac{x^3}{y^2}$ is $y^3(x+1)^2 = \frac{x^6}{a} + \frac{2}{b}x^5 + \frac{1}{c}x^4 + k$, then the value of $(a+b+c)$ is

A) 5 B) 10 C) 15 D) 20

53. In a building programme the event that all the materials will be delivered at the correct time is M and the event that the building programme will be completed on time is F. If $P(M) = 0.8$, $P(M \cap F) = 0.65$ and $P(F) = 0.7$, then the probability that the building programme will be completed on time if all the materials are not delivered at the correct time is

A) $\frac{1}{4}$ B) $\frac{3}{4}$ C) $\frac{3}{20}$ D) $\frac{1}{20}$

54. Product of the solutions of the equation $|\log_{\sqrt{3}} x - 2| - |\log_3 x - 2| = 2$ is

A) -1 B) 3 C) 2 D) 1

55. Locus of the image of point $(1,2)$ in the line $(x - y + 2) + \lambda(2x - 3y + 1) = 0$, where λ is a real parameter, is

A) $x^2 + y^2 + 10x + 6y - 27 = 0$

B) $x^2 + y^2 - 10x + 6y - 27 - 0$

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C) $x^2+y^2-10x-6y-27=0$

D) $x^2+y^2+10x-6y-27=0$

56. The value of $\dfrac{\int_0^1 \left(1-x^{50}\right)^{100} dx}{\int_0^1 \left(1-x^{50}\right)^{101} dx}$ is equal to

A) $\dfrac{5050}{5051}$ B) $\dfrac{5050}{5049}$ C) $\dfrac{5051}{5050}$ D) $\dfrac{5049}{5051}$

57. The area of triangle formed by the line joining origin to the points of intersection of the line $x\sqrt{5}+2y=3\sqrt{5}$ and the circle $x^2+y^2=10$ is

A) 3 B) 4 C) 5 D) 6

58. Let A and B be two points on the lines $L_1: \dfrac{x-1}{1}=\dfrac{y-2}{-1}=\dfrac{z-1}{1}$ and $L_2: \dfrac{x-2}{2}=\dfrac{y+1}{1}=\dfrac{z+1}{2}$ respectively, which are nearest to each other, then the distance between A and B is____

A) $\dfrac{1}{\sqrt{6}}$ B) $\dfrac{3}{2}$ C) $\dfrac{3\sqrt{2}}{2}$ D) $3\sqrt{2}$

59. If $f: R \to R$ be a differentiable function such that $f(x+2y)=f(x)+f(2y)+4xy \ \forall\ x,\ y \in R$, then

A) $f'(1)=f'(0)+1$ B) $f'(1)=f'(0)-1$

C) $f'(0)=f'(1)+2$ D) $f'(0)=f'(1)-2$

60. A family consists of a grandfather, m sons and daughters and $2n$ grandchildren. They are to be seated in a row for dinner. The grandchildren wish to occupy the n seats at each end and the grandfather refuses to have a grandchild on either side of him. The number of ways in which the family can be seated for dinner is

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A) $(2n-1)!m!(m-1)$ B) $(2n)!(m-1)!(m-1)$

C) $(2n)!m!(m-1)$ D) $(2n)!m!m$

61. If $f(x)=|\sin^{-1}x+\tan^{-1}x|+|\cos^{-1}x+\cot^{-1}x|+|\operatorname{cosec}^{-1}x+\sec^{-1}x|$, then the product of maximum and minimum values of $f(x)$

A) $\frac{3\pi^2}{2}$ B) $\frac{5\pi^2}{2}$ C) $\frac{7\pi^2}{2}$ D) $\frac{9\pi^2}{2}$

62. A polynomial function $f(x)$ satisfies the condition $f(x+1)=f(x)+2x+1; f(0)=1$, then the area enclosed by the curve and the pair of tangents from origin is

A) $\frac{3}{2}$ B) $\frac{2}{3}$ C) $\frac{3}{4}$ D) 1

63. If A_1 denotes area of the region bounded by the curves $C_1: y=(x-1)e^x$, tangent to C_1 at $(1,0)$ and $y-$ axis and A_2 denotes area of the region bounded by C_1 and co-ordinate axes in fourth quadrant, then

A) $2A_1+A_2=1$ B) $A_1+2A_2=3$

C) $2A_1+A_2=2$ D) $A_1+2A_2=4$

64. If vertex and focus of hyperbola are $(2,3)$ and $(6,3)$ respectively and eccentricity 'e' of the hyperbola is 2, then the equation of hyperbola is

A) $\frac{(x+1)^2}{16}-\frac{(y-3)^2}{48}=1$ B) $\frac{(x+2)^2}{9}-\frac{(y-3)^2}{27}=1$

C) $\frac{(x+2)^2}{16}-\frac{(y-3)^2}{48}=1$ D) $\frac{(x+2)^2}{6}-\frac{(y-3)^2}{27}=1$

65. If $S_n=\sum_{r=1}^{n}t_r=\frac{1}{6}n(2n^2+9n+13)$, then $\sum_{r=1}^{100}r.\sqrt{t_r}$ equals

A) 333300 B) 343400 C) 344300 D) 443300

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66. A bag contains 10 white 3 black balls. Balls are drawn one by one without replacement till all black balls are drawn. The probability that the procedure of drawing balls will come to an end at the 7th draw is equal to $\frac{a}{b}$, where a and b are coprime natural numbers, then the number of divisors of $'a'$ is___

A) 2 B) 3 C) 4 D) 5

67. Let $A=\{2,3,4\}$ and $B=\{8,9,12\}$, then the number of elements in the relation

$R=\{((a_1,b_1),(a_2,b_2))\in(A\times B,A\times B):a_1 \text{ divides } b_2 \text{ and } a_2 \text{ divides } b_1\}$

is

A) 18 B) 24 C) 12 D) 36

68. If one of the diameters of circle $C_1: x^2+y^2-2x-6y+6=0$ is a chord to the circle C_2 with centre $(2,1)$, then radius of the circle C_2 is

A) 3 B) 2 C) $\frac{3}{2}$ D) 1

69. The sum of all the elements of set $X=\{x\in\{1,2,.....,100\}:\text{HCF}(x,24)=1\}$ is

A) 1440 B) 1536 C) 1537 D) 1633

70. If z and ω are two non-zero complex numbers such that $|z\omega|=1$ and $\arg(z)-\arg(\omega)=\frac{\pi}{2}$, then the value of $i\bar{z}\omega$ is

A) i B) 1 C) $-i$ D) -1

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SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

71. Let A be a non-zero matrix of order '2' such that $A+adjA=A^{-1}$, then determinant of $\left(2A^{-1}\right)$ is

72. The value of the expression $\dfrac{1}{\sum_{r=0}^{50}(-2)^r.\left(\dfrac{^{50}C_r}{r+1}\right)}$ is

73. Twice of the area of parallelogram constructed using vectors $\vec{a}=\vec{p}+2\vec{q}$ and $\vec{b}=2\vec{p}+\vec{q}$ as its adjacent sides, where $\vec{p}$ and $\vec{q}$ are unit vectors containing an angle of $30°$, is

74. Suppose $y=f(x)$ and $y=g(x)$ are two functions whose graphs intersect at three points $(0,4)$, $(2,2)$ and $(4,0)$ with $f(x)>g(x)$ for $0<x<2$ and $f(x)<g(x)$ for $2<x<4$. If $\int_0^4\left(f(x)-g(x)\right)dx=10$, $\int_2^4\left(g(x)-f(x)\right)dx=5$, then the area between two curves for $0<x<2$, is

75. If the foot of perpendicular from point $(4,3,8)$ on the line $L_1:\dfrac{x-a}{l}=\dfrac{y-2}{3}=\dfrac{z-b}{4};\ l\neq 0$ is $(3,5,7)$, then the reciprocal of the square of the shortest distance between the line L_1 and the line $L_2:\dfrac{x-2}{3}=\dfrac{y-4}{4}=\dfrac{z-5}{5}$ is equal to

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